B. Platforms

time limit per test: 2 seconds
memory limit per test: 64 megabytes

In one one-dimensional world there are n platforms. Platform with index k (platforms are numbered from 1) is a segment with coordinates $[(k-1)m, (k-1)m+l]$, and $l < m$. Grasshopper Bob starts to jump along the platforms from point 0, with each jump he moves exactly d units right. Find out the coordinate of the point, where Bob will fall down. The grasshopper falls down, if he finds himself not on the platform, but if he finds himself on the edge of the platform, he doesn't fall down.

Input

The first input line contains 4 integer numbers n, d, m, l $(1 \le n, d, m, l \le 10^6, l < m)$ — respectively: amount of platforms, length of the grasshopper Bob's jump, and numbers m and l needed to find coordinates of the k-th platform: $[(k-1)m, (k-1)m+l]$.

Output

Output the coordinates of the point, where the grosshopper will fall down. Don't forget that if Bob finds himself on the platform edge, he doesn't fall down.

Examples

input Copy

```
2 2 5 3
```

output Copy

```
4
```

input Copy

```
5 4 11 8
```

output Copy

```
20
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 18 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371086088	Apr/14/2026 09:20	Accepted
371085806	Apr/14/2026 09:17	Accepted
371085201	Apr/14/2026 09:11	Accepted

371085087	Apr/14/2026 09:09	**Accepted**
371084720	Apr/14/2026 09:06	Wrong answer on test 2
371084587	Apr/14/2026 09:04	Wrong answer on test 2

→ Problem tags

brute force | math | *1700

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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